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Impact of medical school on quality of life and mental health in Brazil: a cross-sectional comparative study

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ABSTRACT

Objectives This study aimed to compare the perceptions of quality of life (QoL) and mental health among medical students and their peers in other university courses in Brazil through a cross-sectional analysis. We hypothesised that medical students face greater psychological challenges due to the demanding nature of their academic workload. Previous studies have indicated that medical training is associated with a decline in empathy and an increase in stress and anxiety, particularly during the clinical phase, when students face greater exposure to patient care and emotionally demanding experiences. These factors contribute to decreased psychological well-being, highlighting the need for targeted interventions in medical education. To address these challenges, this study investigates the specific impact of medical education on students' mental health and QoL, aiming to identify potential structural changes that could mitigate these negative outcomes.

Design A cross-sectional study was conducted in private higher education institutions in Brazil.

Setting Data were collected online via the QuestionPro platform in August 2024, encompassing 32 units located across 14 states and 4 geographic regions in Brazil.

Participants The sample included 10 844 students, 33.7% of whom were enrolled in medicine and 66.3% in other fields (administration, agronomy, agribusiness, systems analysis and development, architecture, architecture and urbanism, biomedicine, computer science, accounting, economics, social communication/advertising and propaganda, law, physical education, nursing, civil engineering, computer engineering, production engineering, electrical engineering, mechatronics engineering, aesthetics and cosmics, pharmacy, physics, physiotherapy, speech, speech therapy, environmental management, commercial management, human resources management, financial management, history, Portuguese-English literature, logistics, marketing, mathematics, veterinary medicine, nutrition, dentistry, pedagogy, management processes, psychology, advertising and propaganda, computer networks, social work, information systems and theology). The inclusion criterion was as follows: regularly enrolled students. The exclusion criteria were refusal to provide consent and incomplete questionnaires.

The margin of error, calculated as 0.9 percentage points at a 95% CI, was based on a population of 74 684 students enrolled in the private institutions participating in the study.

Primary and secondary outcomes Primary outcomes included QoL assessment via the WHOQOL-Brief Vbreif version and a customised questionnaire developed by the researchers. This questionnaire evaluated variables such as inclusion, accessibility, sports practices, adaptation and satisfaction with undergraduate training. Secondary outcomes assessed factors such as risky substance use, academic satisfaction and mental health conditions, including symptoms of anxiety and depression.

Results Medical students exhibited significant declines in physical QoL (64.5–57.1, p<0.001) and psychological QoL (63.1–53.7, p<0.001) when students at the beginning of the course were compared with those near graduation. Additionally, symptoms of anxiety and depression were more prevalent among medical students, with 61.8% reporting current anxiety and 28.1% reporting current depression. The use of substances, including antidepressants and cognitive enhancers, such as 'smart drugs', was also greater among medical students than among their peers from other courses (21.4% and 11.2%,

STRENGTHS AND LIMITATIONS OF THIS STUDY

- ⇒ This study analysed a large sample of 10 844 students from 32 private higher education institutions across 14 states in Brazil, providing diverse and robust data.
- ⇒ The combined use of the WHO Quality of Life-Brief Vbreif version, a validated quality-of-life assessment tool, and a tailored questionnaire enhanced the study's methodological rigor.
- ⇒ Online data collection allowed wide accessibility but may have introduced selection bias related to internet access and voluntary participation.
- ⇒ The cross-sectional design captured a snapshot of students' experiences but limited the ability to infer causal relationships.
- ⇒ The study focused solely on private institutions, which restricts the generalisability of the findings to public universities with different socioeconomic and academic contexts.

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respectively). These findings underscore the urgency of implementing targeted mental health interventions and institutional reforms to increase the QoL of medical students.

Conclusions This study reveals that medical students in Brazil experience a significant decline in QoL and mental health, particularly in the psychological and physical domains, as they progress through their academic journey. The higher prevalence of anxiety, depression and substance use among medical students than among peers in other fields underscores the intense emotional and academic pressures within medical education. These statistically significant findings highlight the critical need for comprehensive mental health support, curriculum adjustments to promote well-being and inclusive institutional policies. Implementing such measures is vital to enhance student welfare and foster resilient future healthcare professionals. Longitudinal research is necessary to assess the long-term impact of these interventions and to further explore systemic inequalities affecting student well-being.

INTRODUCTION

The quality of life (QoL) of undergraduate students has become a pressing concern in the Brazilian academic environment, particularly in high-demand medical courses.[1 2]

Medical students face unique challenges that distinguish their academic experiences from those of peers in other fields. These challenges are amplified by the rigorous requirements of the National Curriculum Guidelines (*Diretrizes Curriculares Nacionais, DCNs*) in Brazil, which mandate a minimum workload of 7200 hours—nearly double the average for most undergraduate programmes in Brazil.[3] While this extensive training is designed to ensure high professional standards, it also contributes to emotional exhaustion, decreased empathy and long-term mental health issues among future physicians.[4 5]

In addition to academic stressors, the psychological burden on medical students is reflected in alarming mental health statistics. In Brazil, approximately 31.3% of medical students in Brazil report symptoms of depression, while 70.8% experience anxiety, which is significantly greater than that reported in other undergraduate programmes.[1 2] Studies also indicate a progressive decline in empathy throughout medical training, particularly during later years and residency.[6] This phenomenon is closely linked to the stressors of the clinical learning environment, such as exposure to emotionally challenging situations, high academic demands and the influence of negative role models.[4 5] Additionally, the 'hidden curriculum'—unspoken institutional norms and values—often reinforces a focus on technical skills at the expense of humanistic approaches, further exacerbating emotional distress.[7]

These mental health challenges not only impact students' personal well-being but also have broader consequences for the healthcare system; burnout and psychological distress in medical students have been associated with decreased quality of patient care, increased risk of medical errors and diminished professional satisfaction.[8 9]

Given the stigma surrounding mental health in medical culture, coupled with limited time and resources, many students hesitate to seek psychological support, further worsening their distress.[10]

Recognising the multifaceted nature of QoL—encompassing physical, psychological and social dimensions—this study employs the WHOQOL-Brief Version (BREF), a validated tool developed by the WHO,[11] complemented by a lifestyle assessment assessing inclusion, accessibility and satisfaction with undergraduate training. By analysing these factors, we aimed to identify key determinants affecting QoL and mental health among medical students compared with their peers in other academic fields.

This cross-sectional analysis, conducted in private higher education institutions in Brazil, not only examines differences in QoL and mental health across disciplines but also investigates the structural factors contributing to these disparities. By identifying areas for intervention, this research seeks to inform policy changes and institutional reforms that promote healthier academic environments and cultivate mentally resilient professionals, which is essential for a more humanised and effective healthcare system.

MATERIALS AND METHODS

Patient and public involvement statement

This study involved students from private higher education institutions in Brazil. Participation was voluntary, with all participants accepting the Informed Consent Form (Termo de Consentimento Livre e Esclarecido - TCLE) and the study was approved by the CEP under protocol number 79618224.7.0000.9667.

Participants and study design

This cross-sectional study aimed to assess the QoL of undergraduate students from 32 private institutions in Brazil, distributed across 14 states and 4 geographic regions, all belonging to the same higher education group. Detailed data can be found in online supplemental table 1. All the students enrolled in any of the group's colleges were eligible to participate in the study, which was extensively publicised through all the available communication channels, both online and offline. The flow of student participation in the questionnaire was as follows: a total of 74 684 students were impacted, resulting in 37 604 accesses to the questionnaire. Of these, 18 324 students started the questionnaire and 10 844 students completed it, resulting in an engagement rate of 14.5%. Among those who completed the questionnaire, 3659 were medical students (33.7%) and 7185 were from other fields (66.3%). These data are essential for understanding the engagement and response rates throughout the study.

Data were collected from 29 July 2024 to 3 September 2024. These institutions offered 45 undergraduate programmes with different learning formats: 4% in online education/distance learning, 79.6% in fully inperson classes and 16.4% in hybrid models (partially inperson and partially online education).

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A non-probabilistic intentional sampling strategy was used to maximise representation across fields of study and educational formats while focusing on the specific context of private higher education institutions. The inclusion criteria required the students to be regularly enrolled, whereas the exclusion criteria included refusal to provide informed consent and incomplete questionnaires. This approach ensured a diverse sample, enabling comparative analysis between medical students and their peers. The margin of error, calculated as 0.9 percentage points at a 95% CI, was based on a population of 74 684 students enrolled in the private institutions participating in the study.

Data collection

Data were collected through an online survey on the QuestionPro platform from 29 July 2024 to 3 September 2024. Recruitment was conducted via closed student social media groups, with periodic reminders to maximise participation. A trained research team ensured data consistency and validity via standardised protocols. The structured questionnaire covered demographic variables (age, sex, socioeconomic level), lifestyle factors (physical activity, diet, sleep, stress management) and mental health history (prior diagnoses and use of psychotherapeutic or pharmacological interventions). This approach ensured diverse student profiles and voluntary participation.

Instruments used

To assess the factors influencing QoL and mental health comprehensively, the following instruments were used:

Quality of life (WHOQOL-Bref)

The WHOQOL-BREF, developed by the WHO, was used to measure QoL. This questionnaire has been widely validated and highly reliable across diverse populations and cultural contexts, making it particularly suitable for studies with heterogeneous samples, such as university students.[2] The validity and reliability of the WHOQOL-BREF reinforce the accuracy of the findings, ensuring that the instrument rigorously assesses participants' QoL perceptions across multiple dimensions. In this study, scores from medical students and other disciplines were analysed, allowing for detailed comparisons between groups.[11]

WHOQOL-BREF questionnaire

The WHOQOL-BREF is an abbreviated version of the WHOQOL-100, developed by the WHO to assess QoL broadly and multidimensionally.[12] Composed of 26 questions, the instrument covers four main domains that have been validated for different cultures and populations, offering reliable and consistent results for analysing individuals' perceptions of well-being. We used a Brazilian Portuguese translation of the WHOQOL-BREF, as validated by the WHOQOL Centre for Brazil.[12] The domains assessed are as follows:

Physical: This domain includes questions on pain and discomfort, energy and fatigue, sleep and rest, capacity to perform daily activities and dependence on medications or treatments. It assesses how these factors affect physical functioning and energy levels, which is essential for identifying possible challenges in students' routines.

Psychological: This domain encompasses the assessment of positive and negative feelings, memory, concentration, self-image and emotional states such as anxiety and depression. It provides a deeper view of students' mental and emotional well-being, identifying vulnerabilities that can impact QoL and academic performance.

Social relationships: This domain includes questions about the quality of personal relationships, social support received and satisfaction with one's sexual life. It reflects the level of perceived social support and the quality of interpersonal bonds, which are fundamental for students' emotional balance and sense of belonging.

Environment: This domain examines aspects such as physical safety, home environment conditions, financial resources, leisure opportunities, transportation and the quality of health services. It considers external conditions that influence QoL and can directly impact mental health and emotional stability in unfavourable cases.

The WHOQOL-BREF allows the collection of robust and comparable data due to its reliable structure, which has been validated across various populations. These domains help map areas influencing the global perception of QoL and provide insights that can support specific interventions in the academic environment.

Each of the 26 questions in the WHOQOL-BREF has response options on a scale from 1 to 5. The score for each domain is calculated by summing the values assigned to the responses of the corresponding questions and then transforming these scores into a scale from 0 to 100, where higher scores indicate a better perception of QoL.[11] To calculate the domain scores, first, the points from the responses of each item within the domain are summed.[11] Then, a linear transformation is applied to convert these scores into a scale from 0 to 100. Additionally, the first two items of the questionnaire provide a global assessment of overall QoL and general health but are not included in the domain calculations.[11] This scoring method allows for standardised comparisons between different populations and cultural contexts, ensuring the validity and reliability of the results obtained.[11]

Lifestyle assessment

In addition to the WHOQOL-BREF, 42 questions address aspects related to student profiles, inclusion and accessibility, QoL, sports practices, adaptation and satisfaction with the undergraduate degree. The following components were included:

Student profile: students' profiles, including demographic aspects, educational backgrounds, and interests, were evaluated.

Inclusion and Accessibility: students were asked to self-identify as typical learners, neurodivergent individuals, people with disabilities or those with learning disorders. This component aimed to capture how these identities

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impacted their academic experiences, perceptions of institutional support and access to necessary resources.

Quality of Life: assess participants' overall well-being, including emotional, physical and social aspects contributing to their QoL. This assessment is crucial for identifying areas needing intervention and additional support, especially in a challenging academic context.

Sports Practices: analysed the frequency and quality of physical activities performed by students, aiming to understand how these practices influence participants' mental and physical health. Regular physical activity is recognised for its ability to reduce stress and promote well-being.

Adaptation and Satisfaction with undergraduate training: this component addresses students' perceptions of the adequacy of the training received and their satisfaction with teaching methods, infrastructure and academic support. Data analysis will help identify potential areas for improvement in the curriculum and the pedagogical approach of institutions.

Risky alcohol and drug use: harmful use of substances was evaluated on the basis of the WHO criteria, which classify individuals reporting frequent and excessive consumption of alcohol and other drugs as at-risk users. This information is crucial for understanding the correlation of substance use with mental health and QoL among students.

Statistical analysis

The data were analysed via SPSS software, V.29.9. Continuous variables are presented as the means±SDs and categorical variables are presented as percentages with 95% CIs. For univariate analysis, t-tests and χ^2 tests were performed. This detailed methodology ensures the transparency and reproducibility of the study, providing comprehensive information on the instruments and procedures used for data collection and analysis.

RESULTS

Next, we analyse students' QoL, mental health and academic experiences, including demographic and socioeconomic profiles, academic dedication and living conditions. The key findings include mental health disparities, substance use patterns and course satisfaction. Finally, we used WHOQOL-BREF data to highlight differences between medical and non-medical students across QoL domains.

Sample Profile

The sample comprised 10 844 undergraduate students from private higher education institutions in Brazil, 33.7% of whom were enrolled in medical courses and 66.3% in other fields, such as the humanities, health sciences (excluding medicine) and exact sciences. The geographical distribution highlighted regional disparities, with most participants concentrated in the Southeast (34.6%), Northeast (34%) and North (27.6%) regions. The South represented only 3.6% of the sample, and no data were collected from the Middle West region because of the absence of participating institutions.

The gender identity distribution revealed that 69% of the participants identified as female, 29.7% as male and 1.3% as non-binary, agender or other identities. This demographic composition reflects broader trends in higher education, where female students represent a growing majority, particularly in health-related fields.[13] The sample also exhibited racial diversity: 47.3% of the participants identified as white, 41.2% as mixed race, 8.4% as black, 2.6% as Asian and 0.5% as Brazilian Natives.[8] Statistical analysis revealed significant racial disparities across fields: among medical students, 56.8% identified as white, whereas only 36.1% and 4.4% identified as mixed race and black, respectively. In other fields, the proportion of black students reached 10.5% and that of mixed-race students reached 43.8% (χ^2=2 51 064; p<0001). These findings point to structural barriers in accessing medical education, particularly for historically under-represented racial groups.[14]

Age disparities were also notable. While 39.6% of medical students were aged 25 years or older, only 27.7% of students in other fields fell into this category (χ^2 = 221.744; p<0.001). Additionally, 30.2% of medical students held prior undergraduate degrees, compared with 11.1% of their peers (χ^2 = 614.244; p<0.001). These patterns likely reflect the competitive nature of medical school admissions, which often require extensive preparation (table 1).

Dedication, socioeconomic profile, mobility and housing conditions of students

Socioeconomic disparities between medical students and their peers in other fields were pronounced, particularly with respect to family income, mobility and housing conditions.

Exclusive dedication to studies and family income

A total of 75.1% of medical students reported focusing entirely on their education, whereas only 34.6% of students in other fields reported balancing work and studies (χ^2=1 591 739; p<0.001). This high dedication rate is closely related to the financial support of medical students provided by their families, whose average family income is R$ 22 385.20 (approximately US$4600), nearly double the average income of students in other fields, which is R$ 11 500.70 (approximately US$2400)—table 1.

With respect to education, 26.8% of the students moved specifically for academic purposes; this figure rose to 51.1% among medical students, reflecting a greater willingness to relocate to pursue medical training.

Mental health: depression and anxiety

The findings reveal significant mental health challenges among students, with marked differences between medical students and their peers in other fields. Overall, 24.7% of the students reported having received a formal diagnosis

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Table 1 Demographic and socioeconomic profiles of the participants

Characteristic	Total (%)	Medical students (%)	Other fields students (%)
Age in years			
Until 25	68.3%	60.4%	72.3%
26–35	18.3%	23.2%	15.8%
36–45	9.7%	13.5%	7.8%
46–55	3.1%	2.7%	3.3%
56+	0.6%	0.2%	0.8%
Household income			
Up to R$ 5000	43.1%	24.6%	52.5%
Between R$ 5001 and R$ 10 000	21.3%	16.2%	23.8%
Between R$ 10 001 and R$ 15 000	10.0%	12.7%	8.6%
Between R$ 15 001 and R$ 20 000	6.9%	11.8%	4.4%
Between R$ 20 001 and R$ 25 000	4.0%	7.4%	2.3%
Between R$ 25 001 and R$ 30 000	3.4%	6.7%	1.8%
Between R$ 30 001 and R$ 35 000	2.3%	4.2%	1.4%
Between R$ 35 001 and R$ 40 000	1.6%	3.2%	0.8%
Between R$ 40 001 and R$ 45 000	0.8%	1.5%	0.5%
Between R$ 45 001 and R$ 50 000	1.0%	2.0%	0.5%
Between R$ 50 001 and R$ 55 000	0.7%	1.3%	0.4%
Between R$ 55 001 and R$ 60 000	0.5%	0.7%	0.3%
Between R$ 60 001 and R$ 65 000	0.3%	0.5%	0.2%
Between R$ 65 001 and R$ 70 000	0.4%	0.8%	0.2%
Between R$ 70 001 and R$ 75 000	0.3%	0.5%	0.2%
Between R$ 75 001 and R$ 80 000	0.2%	0.4%	0.1%
Between R$ 80 001 and R$ 85 000	0.3%	0.5%	0.1%
Between R$ 85 001 and R$ 90 000	0.2%	0.4%	0.1%
Between R$ 90 001 and R$ 95 000	0.1%	0.3%	0.1%
Between R$ 95 001 and R$ 100 000	0.3%	0.4%	0.3%
Between R$ 100 001 and R$ 105 000	0.2%	0.5%	0.1%
Between R$ 105 001 and R$ 110 000	0.2%	0.2%	0.1%
Between R$ 110 001 and R$ 115 000	0.1%	0.1%	0.1%
Between R$ 115 001 and R$ 120 000	0.2%	0.3%	0.2%
More than R$ 120,000	1.5%	2.9%	0.9%
Has a previous degree			
Yes	17.5%	30.2%	11.1%
No	82.5%	69.8%	88.9%

This table presents data on participants' age range, family income and prior educational background. It includes a breakdown of these characteristics by field of study (medicine vs. other fields).

of a psychiatric disorder such as depression or anxiety. In contrast, when reporting self-reported symptoms without formal evaluation, 45.8% of the participants reported having experienced depressive symptoms, whereas 70.8% reported experiencing symptoms of anxiety at some point in their lives.

DEPRESSION

In general, 45.8% of the students reported experiencing depressive symptoms at some point in their lives. Among medical students, 28.1% reported current symptoms of depression, which was slightly lower than the 31.2% reported by students from other fields. Medical students were also more likely to be undergoing treatment (9.5%) than their peers were (6.7%). However, a substantial proportion of the students in both groups—16.4% among medical students and 21.1% among students in

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other fields—reported experiencing depressive symptoms without seeking specialist care.

Anxiety

Anxiety symptoms were more prevalent than depression symptoms were, with 70.8% of all the students reporting having experienced symptoms at some point. Among medical students, 61.8% reported current symptoms, similar to the 62.3% reported in other fields. Medical students were more likely to be in treatment for anxiety (22.4%) than were students in other fields (14.4%). However, untreated cases remained high in both groups, with 31.8% of medical students and 38.7% of their peers reporting current symptoms without seeking specialist care.

Use of prescribed and non-prescribed substances

The use of psychoactive substances was significantly greater among medical students (29.3%) than among students in other fields (16.9%) (χ^2 = 221.480; p<0.001). Among medical students, the most commonly used substances were antidepressants (21.4%) and cognitive enhancers or 'smart drugs' (11.2%). In contrast, antidepressant use among students in other fields was 13.7%, and 'smart drugs' were used by only 3.3% of the students. Self-medication with psychoactive drugs showed similar patterns in both groups, indicating that medical and non-medical students behave alike in terms of this practice.

Alcohol was the most commonly used non-prescribed substance among both groups (36%). Among medical students, 38.9% reported consuming alcohol, whereas 34.5% reported consuming alcohol in other fields. The overall prevalence of non-prescribed substance use (alcohol, tobacco, marijuana and recreational stimulants) was greater among medical students (41.1%) than among students in other fields (37.9%) (χ^2 = 10.201; p<0.001). Tobacco use was reported by 4.8% of medical students and 5.5% of students in other fields, whereas marijuana use was reported by 4.2% and 4.5%, respectively. When asked about changes in substance use, 47.5% of medical students reported a reduction in their consumption, whereas 17.1% reported an increase. Similar trends were observed among students in other fields, with 47.6% reporting a decrease and 18.7% an increase.

Course satisfaction and specific needs

Satisfaction with the course

Satisfaction with the course was assessed via a 5-point Likert scale (1= 'very dissatisfied,' 5= 'very satisfied'). Among all the students, 68.9% reported satisfaction (scores of 4 or 5). Medical students reported a higher satisfaction rate (72.8%) than did students in other fields (66.9%) (χ^2=75.146; p<0.001).

Time management challenges

Challenges in managing time were evaluated via a 5-point Likert scale (1 = 'very difficult,' 5 = 'very easy'). Overall, 48.8% of the students reported finding it 'difficult' or 'very difficult' to manage their time (scores of 1 or 2). Medical students reported slightly greater difficulty (50.3%) than did students in other fields (48%) (χ^2=32.638; p<0.001).

Academic performance and expectations

Academic performance and expectations were assessed via a 5-point Likert scale (1 = 'never,' 5 = 'always'). In terms of academic performance, medical students had a mean score of 3.71, whereas students in other fields scored 3.59 (t(10 842)=6.215; p<0.001). With respect to academic expectations, medical students also reported higher scores, with a mean of 3.43, compared with 3.33 for students in other fields (t(10 842)=5.541; p<0.001).

Insecurity about professional futures

Professional insecurity was assessed via a 5-point Likert scale (1 = 'never,' 5 = 'always'). The mean insecurity score for medical students was 3.14, whereas for students in other fields, it was 3.40. This difference was statistically significant (t (10 842) =–10.721; p<0.001), indicating that students in other fields experience slightly higher levels of insecurity about their professional futures than medical students do.

Specific conditions and disabilities

Among the participants, 8% identified as individuals with disabilities or neurodiverse conditions. This prevalence was slightly higher among medical students (8.9%) than among students from other fields (7.6%), and this difference was statistically significant (χ^2=5.180; p=0.023). Similarly, the proportion of individuals identifying specifically as neurodiverse was also greater among medical students (4.8%) than among their peers from other fields (3.6%), with this difference being statistically significant (χ^2 = 5671; p=0.017).

Specific conditions, such as dyscalculia, dysgraphia, dyspraxia or attention deficit hyperactivity disorder, were reported by 19.9% of all the students. The prevalence was greater among medical students (21.7%) than among students in other fields (18.9%) (χ^2=11.700; p<0.001).

Analysis of QoL

The analysis of QoL between medical students and students from other fields revealed significant differences in most WHOQOL-BREF domains. Medical students reported significantly higher scores in all domains—physical, psychological, social relationships and environmental—indicating a better perceived QoL in these areas compared with their peers from other fields. However, there were no statistically significant differences in the overall perception of QoL and satisfaction with health between the two groups (table 2).

The QoL perceptions of students from medicine and other fields were compared at the beginning and end of their courses. The results revealed that medical students experienced more pronounced declines across multiple domains, highlighting the impact of their academic journeys on their overall QoL.

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Table 2 Analysis of quality of life and comparisons between groups via t-tests

Domain	Total score (Mean±SD)	Group				
		Course	Total score (Mean±SD)	t-value	P value	Statistically significant
Physical	62.1±17.0	Medicine	62.7±17.3	2.614	0.004	Yes
		Other fields	61.8±16.9			
Psychological	58.7±17.6	Medicine	60.4±17.2	7.137	<0.001	Yes
		Other fields	57.9±17.8			
Social relationships	64.9±20.8	Medicine	66.5±20.2	5.694	<0.001	Yes
		Other fields	64.1±21.0			
Environmental	60.9±17.0	Medicine	64.8±16.2	17.401	<0.001	Yes
		Other fields	58.9±17.0			
Quality of life perception	70.2±23.5	Medicine	70.6±25.0	1.244	0.107	No
		Other fields	70.0±22.7			
Satisfaction with health	59.0±26.6	Medicine	59.5±27.4	1.178	0.119	No
		Other fields	58.8±26.2			

This table summarises the global average scores for the WHOQOL-BREF domains—physical, psychological, social relationships and environmental—across the entire sample. The results are presented separately for medical students and students from other fields. The mean scores and SD are included for each group, along with the t-test values and significance levels (p values). Statistically significant differences (p<0.05) highlight disparities between the groups, with medical students generally reporting lower scores in the physical and psychological domains.

Physical domain

In the physical domain, medical students started their courses with a score of 64.5 (SD=16.83), which declined significantly to 57.1 (SD=18.52) by the end of their studies. Students in other fields showed a smaller reduction, from 64.2 (SD=16.40) at the beginning to 61.7 (SD=17.41) at the end. These findings indicate that medical students experienced a sharper decline in their perception of physical well-being.

Psychological domain

The psychological domain scores followed a similar pattern. Medical students began with a mean score of 63.1 (SD=15.67), which decreased significantly to 53.7 (SD=18.29) by the end of the course. In contrast, students from other fields started with a score of 60.2 (SD=17.47) and experienced a smaller reduction to 57.8 (SD=17.62).

Social relationships domain

For the social relationships domain, medical students began with a score of 68.5 (SD=18.82) and ended with a slightly lower score of 65.7 (SD=17.53). Students in other fields reported scores of 65.8 (SD=20.62) at the beginning and 64.5 (SD=21.61) at the end. These changes were not statistically significant for either group, indicating relative stability in this domain.

Environmental domain

The environmental domain showed that medical students started with a score of 65.7 (SD=14.77), which decreased to 63.7 (SD=14.96) by the final semester. Meanwhile, students from other fields began with a lower score of 58.9 (SD=17.07) but experienced a slight increase to 59.4 (SD=17.01) by the end of their courses.

Overall QoL perception and health satisfaction

The overall QoL perception scores for medical students decreased from 73.5 (SD=23.69) at the beginning of the course to 63.6 (SD=27.68) by the end. The students in other fields started with a score of 72.7 (SD=20.43) and ended with a slightly lower score of 70.2 (SD=24.40).

Similarly, satisfaction with health for medical students decreased from 64.7 (SD=24.81) at the beginning to 50.9 (SD=31.24) at the end. Students from other fields reported scores of 60.6 (SD=25.01) at the start and 58.0 (SD=26.92) by the final semester.

These results indicate that medical students experienced a more pronounced decline in QoL across the physical, psychological and overall perception domains than did students in other fields. While students from other fields showed relative stability in their QoL, the significant reductions observed among medical students underscore the challenges faced throughout their academic journeys.

DISCUSSION

The results of this study highlighted structural inequalities, mental health challenges and significant declines in QoL among medical students compared with their peers in other undergraduate programmes. Below, we discuss the key factors associated with these findings, contextualising them within the relevant literature.

Gender, race and socioeconomic inequalities: significant disparities in gender, race and socioeconomic conditions

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were identified among medical students compared with their peers in other fields. Women comprised 69% of the overall sample, aligning with global trends that show an increasing predominance of female students in health and humanities disciplines.[13] However, this growing representation does not eliminate gender-related challenges. Compared with their male counterparts, female students are often more emotionally vulnerable and face intensified social and academic pressures that contribute to higher rates of anxiety and depression.[15] Despite advancements in educational access, persistent inequalities—such as the unequal distribution of domestic responsibilities and gender-based discrimination—may further exacerbate these vulnerabilities.[13 15] Racial disparities were particularly evident in medical education, where black (4.4%) and mixed-race (10.5%) students were significantly under-represented compared with other fields, where these proportions were notably greater (10.5% and 43.8%, respectively). These differences reflect systemic barriers that hinder access to higher education and medical training in Brazil.[4 13] In addition to structural obstacles, minority students often experience institutional racism, which further compounds their academic and emotional burdens.[14 16]

Socioeconomic disparities were also pronounced, as medical students reported a considerably higher average family income (R$ 22 385.20) than their peers in other fields did (R$ 11 500.70). While this financial advantage allows many to focus exclusively on their studies, it also underscores the socioeconomic inequalities that limit access to medical education for low-income students. These findings highlight the need for inclusive policies—such as affirmative action programmes and financial support—to promote greater equity across undergraduate programmes.[16]

Mental health: anxiety and depression

Mental health was significantly affected across the sample, with 70.8% of the students reporting anxiety symptoms and 45.8% reporting depressive symptoms at some point in their lives. Among medical students, 28.1% reported current symptoms of depression, whereas 31.8% reported untreated anxiety symptoms. These untreated cases highlight the challenges students face in accessing mental healthcare, including stigma and lack of time.[10]

Although medical students reported higher rates of treatment for both anxiety (22.4%) and depression (9.5%) than their peers did, the burden of untreated symptoms remains concerning. The significant prevalence of anxiety and depression among medical students, as observed in this study, aligns with previous findings that highlight the detrimental impact of the clinical learning environment on mental health.[1 2 5] The transition to clinical training is particularly challenging, as students experience an increase in stress, exposure to patient suffering and a reduction in personal well-being.[5 6] Additionally, research has shown that the decline in empathy during this period is not necessarily due to a lack of concern but rather a psychological adaptation to cope with stress, which may paradoxically impair their ability to provide compassionate care.[6]

These findings emphasise that the current medical education model, rather than solely fostering resilience and professional competence, may inadvertently contribute to student distress and burnout. This highlights the need for institutional reforms to create a more supportive learning environment that balances academic demands with mental health support. Addressing this issue is crucial not only for students' well-being but also for ensuring the quality of future healthcare delivery, as emotionally distressed physicians are more likely to experience burnout and reduced empathy in clinical practice.

The observed declines in the psychological domain of QoL among medical students, from 63.1 at the beginning of the course to 53.7 by the last semester, further emphasise the role of emotional burdens in shaping their academic experiences. These findings are consistent with prior research on the unique stressors faced in high-demand fields such as medicine.

Quality of life: impacts of academic and personal challenges

The findings of this study highlight a significant decline in perceived QoL among medical students, particularly in the psychological domain, where sharp reductions were observed. These results align with global studies on the high prevalence of anxiety and depression in this population.[16 17] The decline in psychological well-being and overall QoL is consistent with previous research emphasising the emotional burden of medical training.[1 3] The WHOQOL-BREF scores in this study illustrate a progressive decrease in both psychological and physical well-being, a trend also reported internationally, especially in response to clinical exposure and increasing academic pressure.[2] These stressors are further compounded by structural factors within medical education, such as the prioritisation of technical competence over emotional intelligence, the hidden curriculum and the normalisation of emotional distress.[7 8] These findings reinforce the urgent need for educational institutions to reassess the structural and cultural aspects of medical training that contribute to emotional exhaustion. Internationally, discussions on medical education reform have gained momentum, emphasising the role of psychological support systems, mentorship programmes and more balanced curricula. By contributing to this global discourse, this study advocates for a paradigm shift that values not only academic excellence but also the psychological well-being of future healthcare professionals.[18] To address these challenges, targeted interventions such as regular mindfulness programmes and structured psychological support have shown promising results in fostering emotional resilience in demanding academic environments.[19 20] The decline in the physical domain may be linked to unhealthy lifestyle habits developed during medical training, including inadequate sleep, irregular eating patterns and reduced physical activity, all of which

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directly impact students' well-being.[4 20] These issues are often exacerbated by rigid curricula that offer little flexibility for self-care, a concern highlighted in recent reviews on the challenges faced by students in competitive educational systems.[7 8 21] Although the social and environmental domains remained relatively stable, the discrepancy between medical students and their peers in other fields suggests inequalities in perceived institutional support and adaptation to the academic environment. More inclusive policies, such as measures to increase accessibility for neurodiverse students, could help mitigate these negative effects.[22]

Future implications

This study underscores the necessity of re-evaluating the current medical education framework to address student burnout and enhance psychological well-being. Medical schools should consider curricular modifications that integrate mental health support as a core component of training. Policies such as workload adjustments, structured mentorship programmes and targeted psychological interventions could mitigate the negative effects of medical training on students' mental health. Additionally, these insights contribute to the global conversation on medical education reform, emphasising the importance of cultivating a more human-centred and supportive academic atmosphere. Future research should explore longitudinal interventions and their effectiveness in mitigating the negative effects of medical training on students' mental health. These measures are not only essential for improving student well-being but also crucial for ensuring the long-term quality and safety of healthcare professionals in practice.

Recommendations

Inclusive and equitable policies

Expanding affirmative action programmes and financial aid for low-income and racial minority students. These measures can reduce structural inequalities and increase access to medical education.[22 23]

Targeted wellness programmes

These programmes implement mindfulness-based interventions, stress management workshops and physical activities tailored to medical students' needs, promoting both physical and mental health.[19 20]

Integration of digital tools

Adopting online platforms and apps for psychological support, broadening access to services and reducing barriers such as stigma and time constraints.[21]

Support for neurodiverse students

Establish inclusive academic policies with specialised support for students with neurodiverse conditions or other specific needs, improving their educational experience.[22]

Structured peer support groups

Create peer support groups to foster emotional connections and share coping strategies, strengthening collective resilience and a sense of belonging.[24 25 26]

Limitations and future directions

Limitations

This study has several limitations. First, its cross-sectional design restricts the ability to infer causal relationships, offering only a snapshot of students' experiences rather than changes over time. Second, the data were collected exclusively from private higher education institutions, which limits the generalisability of the findings to public universities, where students may face different socioeconomic and academic pressures. Third, the reliance on self-reported data may introduce recall and reporting biases, potentially influencing the accuracy of the results. Despite these limitations, the study provides valuable insights into the challenges faced by medical students in Brazil, supported by a robust sample size and validated assessment tools. Additionally, although the final sample size was robust and statistically adequate—with a margin of error of 0.9 percentage points at a 95% CI—the overall engagement rate was relatively low. Of the 74 684 students reached, 37 604 accessed the questionnaire link, but only 10 844 completed it, corresponding to an engagement rate of approximately 14.5%. This discrepancy may be attributed to the non-unique nature of the questionnaire link, which was broadly disseminated through digital platforms, possibly resulting in repeated or exploratory access not followed by participation. Nonetheless, the final sample was sufficiently large and diverse to support the analyses, and the distribution of responses enabled valid and meaningful comparisons across groups.

Future work and unanswered questions

Future research should aim to conduct longitudinal studies to better understand how medical education impacts students' QoL and mental health over time. The inclusion of public university students in future analyses could expand the understanding of how institutional contexts shape these outcomes. Additionally, evaluating the effectiveness of specific interventions—such as mental health programmes, workload adjustments and peer support groups—could help address the vulnerabilities identified in this study. Finally, exploring cultural and systemic factors, such as the stigma and accessibility of resources, remains critical for developing inclusive support systems tailored to the diverse needs of students.

CONCLUSION

This study highlights significant disparities in QoL and mental health between medical students and their peers in other fields, emphasising the cumulative toll of high academic demands, emotional stressors and personal sacrifices inherent in medical training. The sharper declines observed in physical and psychological QoL

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among medical students underline the urgent need for targeted interventions, including improved mental health support programmes, workload management strategies and institutional policies that foster healthier academic environments.

By addressing these vulnerabilities, universities can not only improve students' QoL but also prepare more resilient healthcare professionals capable of contributing to a humanised and efficient healthcare system. Future research should explore longitudinal impacts and focus on the development of inclusive policies that ensure equitable access to support services, particularly for underrepresented and vulnerable student groups.

Addressing these vulnerabilities requires tailored interventions, including the promotion of inclusive policies to reduce structural inequalities, wellness programmes to support mental and physical health and accessible mental health services to reduce the prevalence of untreated conditions. Additionally, structured peer support groups could foster emotional resilience and improve social connectedness among students.

By implementing these strategies, institutions can not only improve students' QoL but also prepare more resilient professionals who are better equipped to contribute to a humanised and efficient healthcare system. Future research should explore the longitudinal impacts of these interventions and consider specific strategies to mitigate disparities in underrepresented groups, ensuring equity and sustainability in higher education.

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Ethics approval This study involves human participants and the Research Ethics Committee (Comitê de Ética em Pesquisa - CEP) of the Instituto de Ensino Superior Presidente Tancredo de Almeida Neves (IPTAN) approved this study under protocol number 79618224.7.0000.9667. In Brazil, the ethical review system for research is regulated by Resolution No. 466/2012 of the National Health Council (CNS), which establishes guidelines and regulations for research involving human subjects. According to this resolution, approval by a CEP accredited by the National Research Ethics Commission (CONEP) is valid nationwide, provided that the study complies with the established ethical guidelines. IPTAN is an institution accredited by CONEP, meaning that its ethical approval is recognised throughout the country. Therefore, the approval obtained at IPTAN is valid for all participating institutions in this study, regardless of their geographical location. Additionally, all the institutions involved were informed about the study and agreed to its execution, ensuring compliance with national and international ethical standards. Participants gave informed consent to participate in the study before taking part.

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